December 19th, 2024

UC Berkeley Event Services

TEAM 3 DP FINAL REPORT

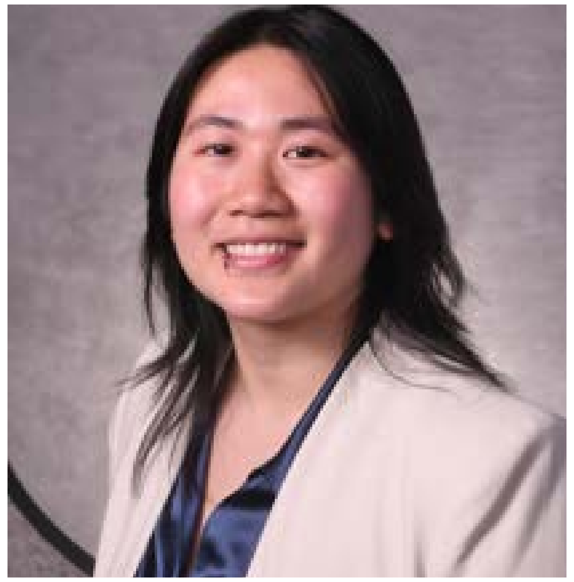
UC Berkeley
Event Services

AGENDA

1. INTRODUCTION
2. EER DIAGRAM
3. RELATIONAL SCHEMA
4. QUERIES
5. NORMALIZATION
6. DISCUSSION & REFLECTION

MEET TEAM 3!

CEO
Ashmita



CCO
Joanne

Project Manager
Jacqueline

UC Berkeley
Event Services

TEAM 3 DATABASE DESIGNERS


Andrew


Christopher


Ella


Ernest


Gursimar


Hannah


Michael


Moritz


Shishira

UC Berkeley Event Services

CLIENT INTRODUCTION

About Them

20+
Years of operation

1000+
Events annually

100+
Employees (students & full-time staff)

Services

Event Planning

Venue Bookings

Student Room Reservations

Catering

PRESENT DATABASE GOALS

Revenue Generation

Implement **dynamic pricing** strategies, prevent **overstocking**, unexpected repairs, **replacement** of equipment to improve financial planning

Strategic Planning

Forecast long-term capacity needs based on trends in venue demand and capability, make informed decisions about pricing, venue improvements, and maintenance

Database Objectives

Operational Efficiency

Utilize **demand projections** in order to optimize staffing during events and **manage resource allocation**, improve service quality via venue availability data

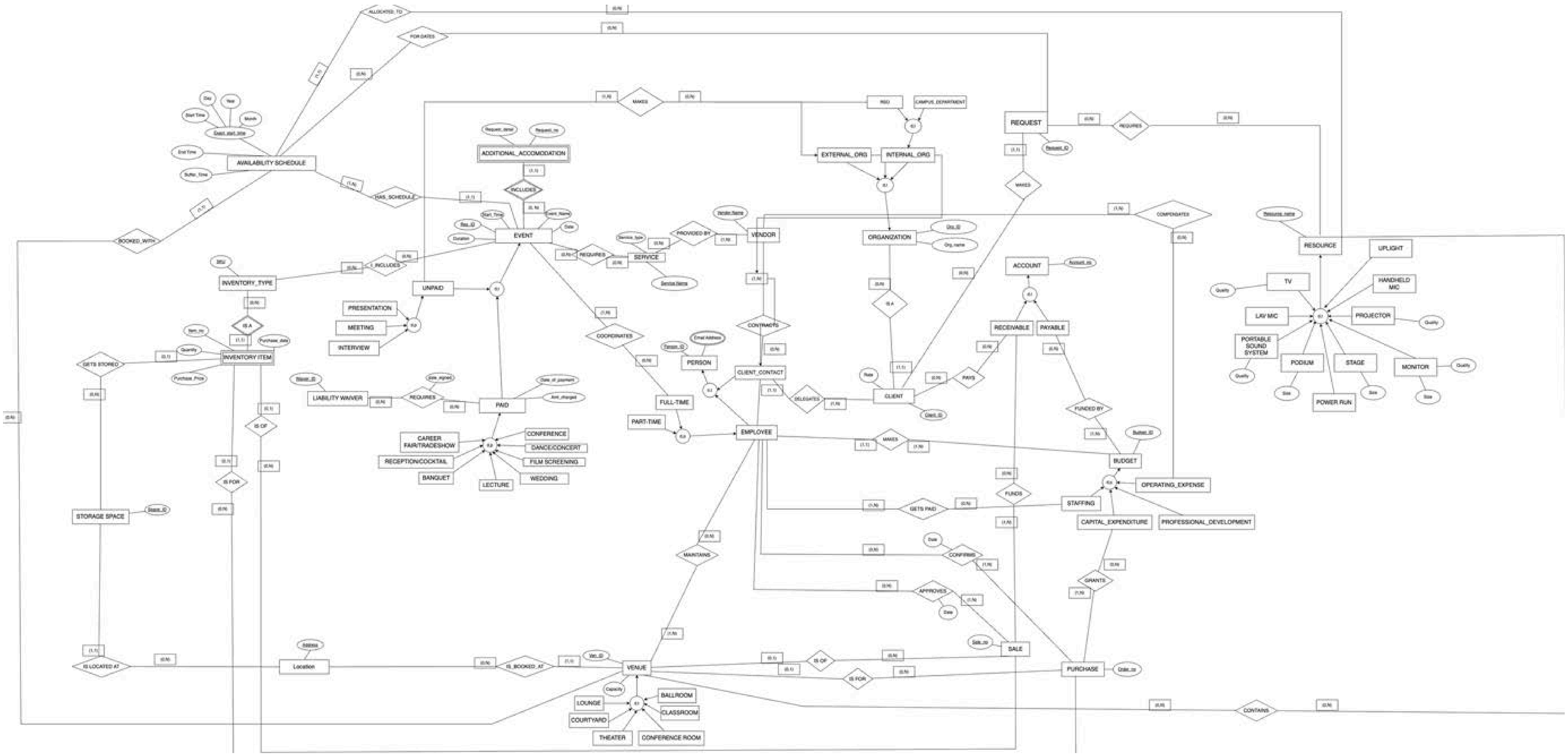
Equipment Management

Track equipment wear and tear using purchase dates and **projected expected lifetimes** to reduce operational risk, proactively plan **maintenance and replacement**

EER DIAGRAM & RELATIONAL SCHEMA

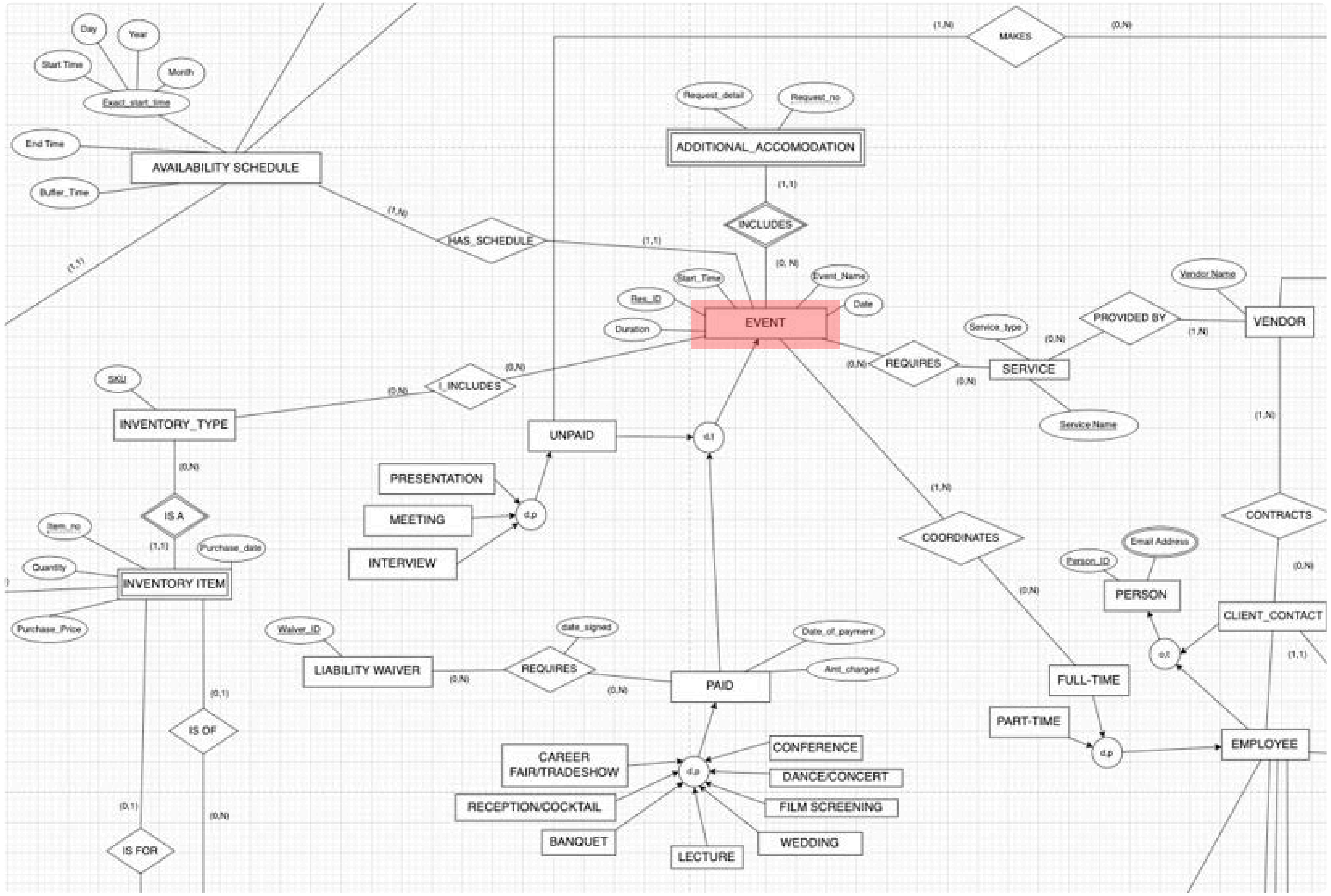
EER DIAGRAM OVERVIEW

UC Berkeley Event Services



UC Berkeley Event Services

ER DIAGRAM- EVENT



Event Entity

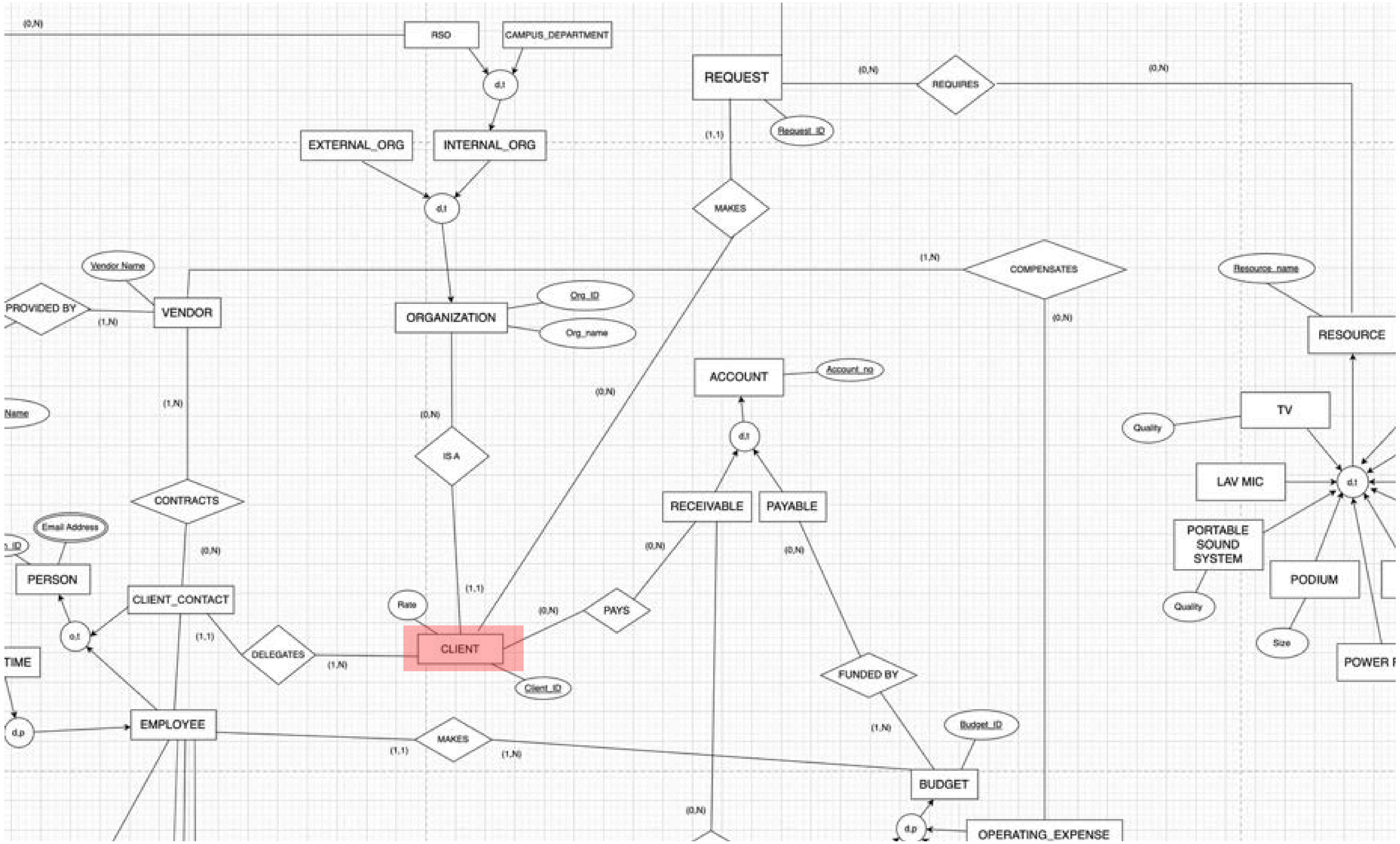
- There are 2 types of events- Paid and Unpaid
- Coordinated by Full-Time Event Services Employees
- Could require additional accommodation requests
- Scheduled based on the availability schedule
- Can include services that are provided by vendors contracted by clients
- Can include inventory needed at the venue

EER DIAGRAM- CLIENT

UC Berkeley Event Services

Client Entity

- Clients are Organizations and are either external or internal organizations
- Delegate contact who contracts vendors
- Makes requests for resources and dates for their event
- Pays Event Service's Accounts Receivable

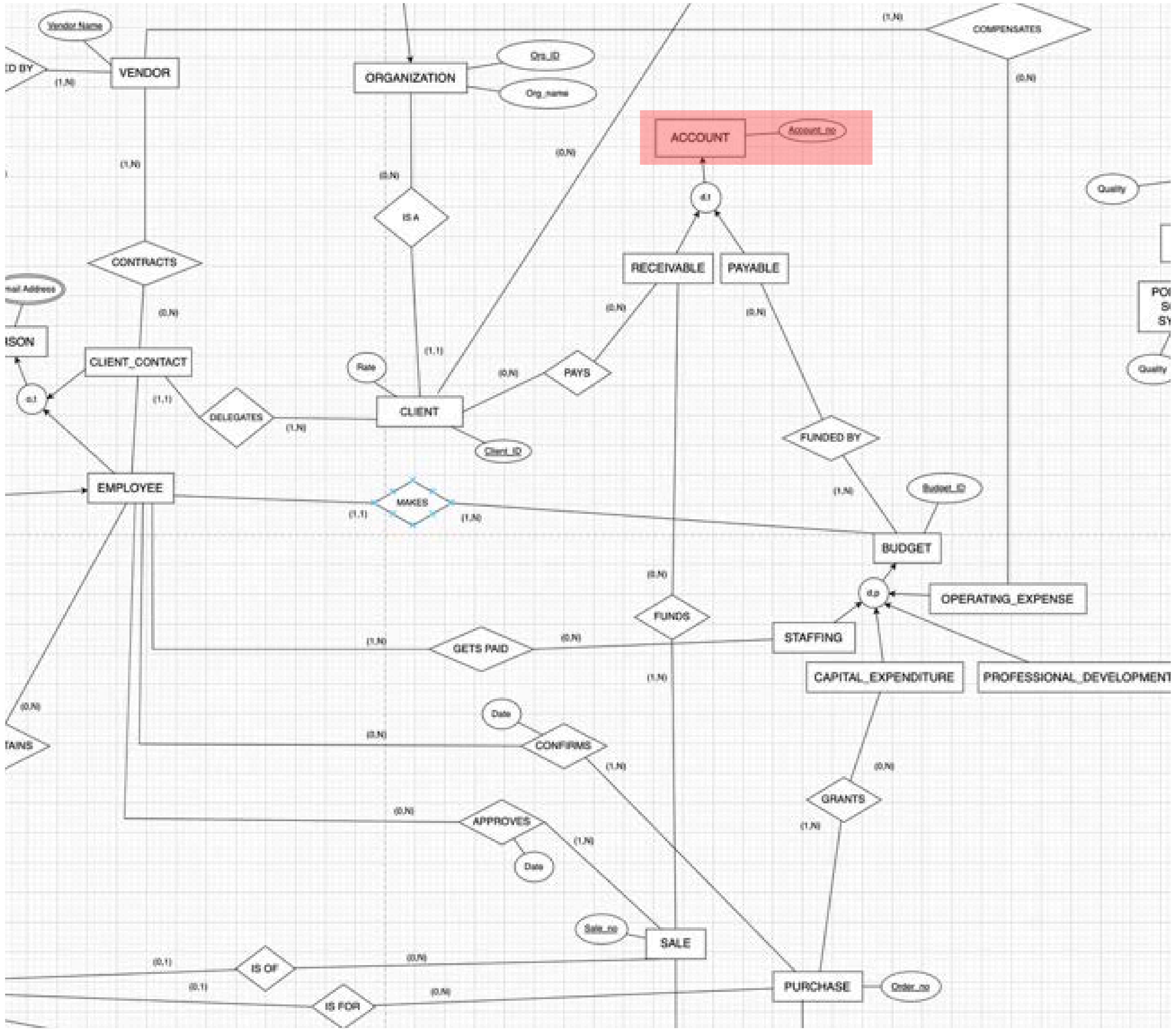


EER DIAGRAM- ACCOUNT

UC Berkeley Event Services

Account Entity

- Account is split into Accounts Receivable and Accounts Payable
- Accounts Receivable receives payments from clients and earnings from purchases at venues
- Accounts Payable is funded by the budget, which is set by employees
- The budget determines compensation for vendors, employee wages, and any purchases made by venues



UC Berkeley Event Services

RELATIONAL SCHEMA

1. EVENT(Res_ID, Duration, Start_Time, Event_Name, Date, | **Ven_ID[21], Date[21], Reservation_Time[21]**)|
    a. UNPAID(Res_ID[1])
        i. PRESENTATION(Res_ID[1a], Presenter_name)
        ii. MEETING(Res_ID[1a], Meeting_topic)
        iii. INTERVIEW(Res_ID[1a], Host_name)
    b. PAID(Res_ID[1], Date_of_payment, Amt_charged, Amt_deposit, Amt_refundable)
        i. CAREER_FAIR_TRADESHOW(Res_ID[1b], Num_of_companies)
        ii. RECEPTION_COCKTAIL(Res_ID[1b], Allergy_accomodation_needed, Host_name)
        iii. BANQUET(Res_ID[1b], Allergy_accomodation_needed)
        iv. LECTURE(Res_ID[1b], Lecturer_name)
        v. WEDDING(Res_ID[1b], Allergy_accomodation_needed)
        vi. FILM_SCREENING(Res_ID[1b], Producer_name)
        vii. DANCE_CONCERT(Res_ID[1b], Performance_type)
        viii. CONFERENCE(Res_ID[1b], Conference_type)
2. ADDITIONAL_ACCOMODATION(Request_no, Res_ID[1], Request_detail)
3. SERVICE(Service_name, Service_type)
4. PERSON(Person_ID, First_name, Last_name)
    a. EMPLOYEE(Person_ID[4], SSN, Start_date, Mailing_address, Bank_account_no, Birthdate, | **Makes_Budget_ID[19]**)
        i. FULL_TIME(Person_ID[4a], Salary)
        ii. PART_TIME(Person_ID[4a], Hourly_wage)
    b. CLIENT_CONTACT(Person_ID[4], Relation_to_client, | **Delegate_of_Client_ID[8]**)
5. EMAIL_ADDRESSES(Person_ID[4], Email_address)
6. LIABILITY_WAVER(Waiver_ID, Date_last_updated)
7. LOCATION(Address, Building_name)
8. CLIENT(Client_ID, Client_name, Rate, Mailing_address, | **Org_ID[11]**)
9. INVENTORY_TYPE(SKU, Type_name, Inventory_expected_lifetime)
10. INVENTORY_ITEM(SKU[9], Item_no, Purchase_price, Purchase_date, | **Gets_stored_Space_ID[20], Inv_Is_of_Sale_no[16], Inv_Is_for_Order_no[17]**)
11. ORGANIZATION(Org_ID, Num_members, Org_name, Tax_ID)
    a. EXTERNAL_ORG(Org_ID[11], Is_nonprofit)
    b. INTERNAL_ORG(Org_ID[11], Has_student_members)
        i. RSO(Org_ID[11b], Advisor_department, Advisor_name)
        ii. CAMPUS_DEPARTMENT(Org_ID[11b], Dept_name)
12. REQUEST(Request_ID, request_date, | **Made_by_Client_ID[8]**)
13. RESOURCE(Resource_name, Resource_expected_lifetime)
    a. TV(Resource_Name[13], Quality, Resolution, Screen_size)
    b. LAV_MIC(Resource_Name[13], Resource_frequency)
    c. PORTABLE_SOUND_SYSTEM(Resource_Name[13], Quality)
    d. PODIUM(Resource_Name[13], Size)
    e. POWER_RUN(Resource_Name[13], Resource_voltage)
    f. STAGE(Resource_Name[13], Size)
    g. MONITOR(Resource_Name[13], Quality, Resolution, Screen_size)
    h. PROJECTOR(Resource_Name[13], Quality, Is_built_in)
    i. HANDHELD_MIC(Resource_Name[13], Resource_frequency)
    j. UPLIGHT(Resource_Name[13], Resource_lumens)
14. VENDOR(Vendor_name, Mailing_address, Cancellation_refundable)
15. VENUE(Ven_ID, Venue_name, ADA_accomodation, Capacity, **Is_booked_at_Address[7], Ven_Is_of_Sale_no[16], Ven_Is_for_Order_no[17]**)
    a. LOUNGE(Ven_ID[15], Kitchen_included)
    b. COURTYARD(Ven_ID[15], Weather_accomodation_plan)
    c. THEATER(Ven_ID[15], Orchestra_pit_included, Stage_lights_included)
    d. CONFERENCE_ROOM(Ven_ID[15], Can_livestream)
    e. CLASSROOM(Ven_ID[15], Chalkboard_or_whiteboard)
    f. BALLROOM(Ven_ID[15], Floor_type)
16. SALE(Sale_no, Dollar_amt, Date_completed)
17. PURCHASE(Order_no, Dollar_amt, Date_completed)

UC Berkeley Event Services

RELATIONAL SCHEMA

18. ACCOUNT(Account_no, Bank_name)
    a. RECEIVABLE(Account_no[18], Internal_funds)
    b. PAYABLE(Account_no[18], Reserved_funds)
19. BUDGET(Budget_ID, Dollar_amt, Number_list_items, Date_set, Percent_accurate)
    a. OPERATING EXPENSE(Budget_ID[19], Intended_start_date, Intended_end_date)
    b. PROFESSIONAL_DEVELOPMENT(Budget_ID[19], Expenditure_date, Start_time, Duration)
    c. CAPITAL EXPENDITURE(Budget_ID[19], Expenditure_date)
    d. STAFFING(Budget_ID[19], Intended_start_date, Intended_end_date)
20. STORAGE SPACE(Space_ID, Capacity, | **Is_located_at_Address**[7])
21. AVAILABILITY_SCHEDULE(Ven_ID[15], Date, Reservation_Time, AfterHours_or_ActiveHours)

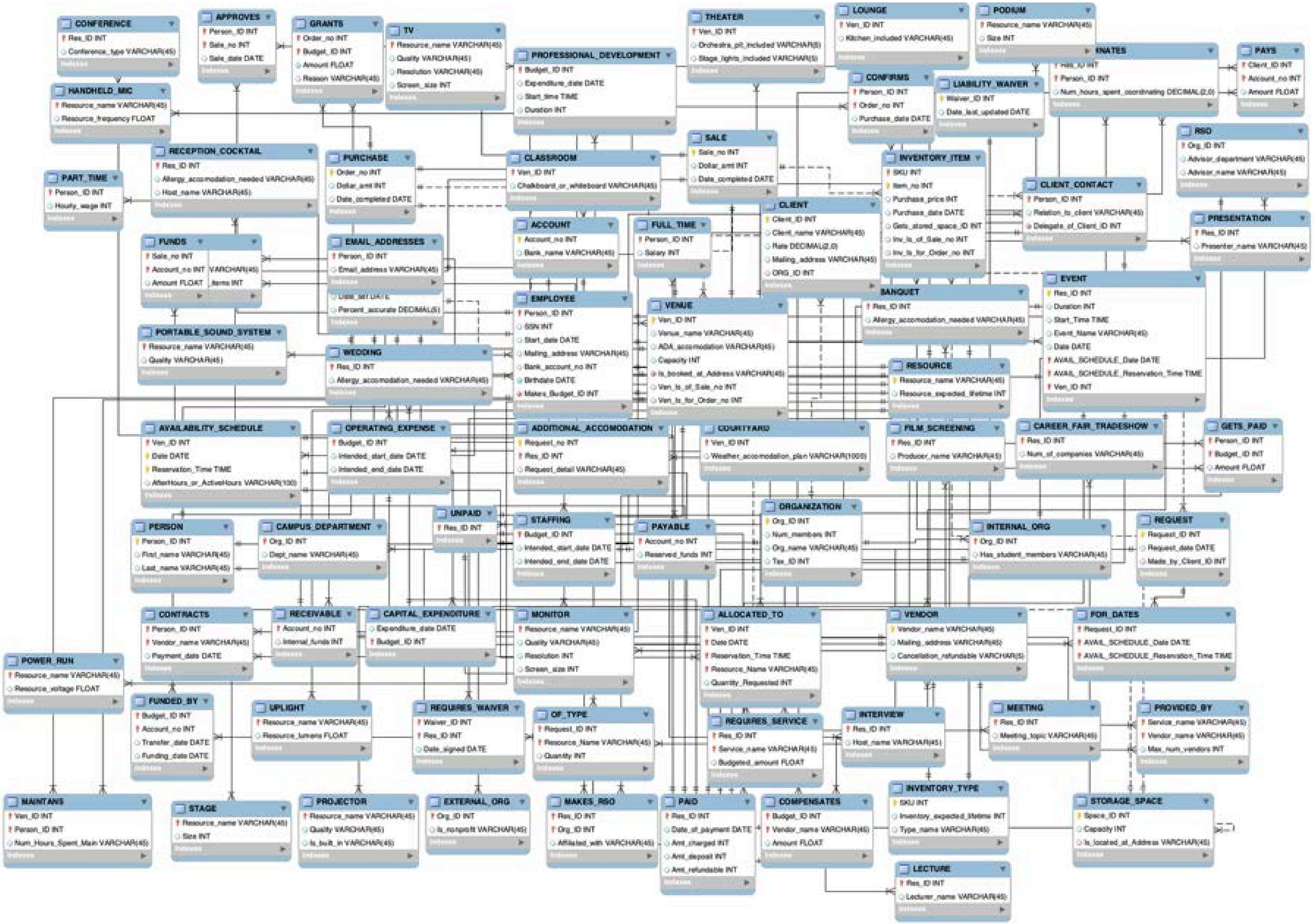
22. COORDINATES(Res_ID[1], Person_ID[4a], Num_Hours_Spent_Coordinating)
23. MAINTAINS(Ven_ID[15], Person_ID[4a], Num_Hours_Spent_Maintaining)
24. FOR_DATES(Request_ID[12], Date[21], Reservation_Time[21])
25. I_INCLUDES(Res_ID[1], SKU[9], Num_items)
26. REQUIRES_SERVICE(Res_ID[1], Service_name[3], Budgeted_amount)
27. PROVIDED_BY(Service_name[3], Vendor_name[14], Max_num_vendors)
28. CONTRACTS(Person_ID[4b], Vendor_name[14], Payment_date)
29. MAKES_RSO(Res_ID[1a], Org_ID[11bi], Affiliated_with)
30. PAYS(Client_ID[8], Account_no[18a], Amount)
31. FUNDED_BY(Budget_ID[19], Account_no[18b], Transfer_date, Funding_date)
32. FUNDS(Sale_no[16], Account_no[18a], Amount)
33. GETS_PAID(Person_ID[4a], Budget_ID[19d], Amount)
34. CONFIRMS(Person_ID[4a], Order_no[17], Purchase_date)
35. APPROVES(Person_ID[4a], Sale_no[16], Sale_date)
36. GRANTS(Order_no[17], Budget_ID[19c], Amount, Reason)
37. CONTAINS(Ven_ID[15], Resource_Name[13], Total_number_of_items)
38. COMPENSATES(Budget_ID[19a], Vendor_name[14], Amount)
39. OF_TYPE(Request_ID[12], Resource_Name[13], Quantity)
40. REQUIRES_WAIVER(Waiver_ID[6], Res_ID[1b], Date_signed)
41. ALLOCATED_TO(Ven_ID[21], Date[21], Reservation_Time[21], Resource_Name, quantity_requested)

UC Berkeley Event Services

Queries

MYSQL WORKBENCH

UC Berkeley Event Services



UC Berkeley
Event Services

Query 1:

Venues Demand Forecasting & Profitability Optimization

UC Berkeley Event Services

QUERY 1: BACKGROUND AND BENEFITS

Revenue Optimization	**1.**	Enabling Event Services to use **dynamic pricing** strategies to maximize revenue
Operational Efficiency	**2.**	Improving **resource allocation** and service quality by understanding demand data
Strategic Planning	**3.**	Inform **long-term decision-making** about capacity, pricing, and infrastructure

QUERY 1: SQL CODE

Demand Forecasting and Profitability Optimization

```
SELECT
    e.Ven_ID AS Venue_ID, v.Venue_name AS Venue_Name, v.ADA_accomodation AS ADA_Accommodation,
    v.Capacity, v.Is_booked_at_Address AS Address, e.Date AS Booking_Date, p.Amt_charged AS Price_Charged,
    e.Duration AS Event_Duration, COALESCE(labor.Total_Labor_Hours, 0) AS Labor_Hours,
    COALESCE(revenue.Total_Revenue, 0) AS Revenue, e.Res_ID

FROM EVENT e
LEFT JOIN VENUE v ON e.Ven_ID = v.Ven_ID
LEFT JOIN PAID p ON e.Res_ID = p.Res_ID
LEFT JOIN (
    SELECT
        Res_ID,
        SUM(Amt_charged) AS Total_Revenue
    FROM PAID
    GROUP BY Res_ID
) revenue ON e.Res_ID = revenue.Res_ID
LEFT JOIN (
    SELECT
        Res_ID,
        SUM(Num_hours_spent_coordinating) AS Total_Labor_Hours
    FROM COORDINATES
    GROUP BY Res_ID
) labor ON e.Res_ID = labor.Res_ID
ORDER BY e.Date;
```

QUERY 1: SQL RESULTS

UC Berkeley Event Services

Demand Forecasting and Profitability Optimization

Venue_ID	Venue_Name	ADA_Accommodation	Capacity	Address	Booking_Date	Price Charged	Event_Duration	Labor_Hours	Revenue	Res_ID
234	Sequoia at Anchor House	No	72	1950 Oxford St, Berkeley, CA 94704	2024-01-08	2010	4	93	2010	1037
84	Kerr Room	No	70	2495 Bancroft Way, Berkeley, CA 94720	2024-11-23	4000	8	96	4000	2973
77	Anna Head Alumnae Room	Yes	379	2537 Haste St, Berkeley, CA 94720	2024-10-23	3539	8	95	3539	2339
77	Anna Head Alumnae Room	Yes	379	2537 Haste St, Berkeley, CA 94720	2024-08-10	3412	8	94	3412	2837
235	Courtyard at Anchor House	Yes	48	1950 Oxford St, Berkeley, CA 94704	2024-09-28	3400	5	96	3400	1724
77	Anna Head Alumnae Room	Yes	379	2537 Haste St, Berkeley, CA 94720	2024-10-17	3375	8	88	3375	2346
236	The Venue at Anchor House	No	48	1950 Oxford St, Berkeley, CA 94704	2024-03-12	3361	8	100	3361	1123
102	Tilden Room	Yes	48	2495 Bancroft Way, Berkeley, CA 94720	2024-08-06	3262	7	95	3262	2785
90	Executive Dining Room	Yes	50	2601 Warring St, Berkeley, CA 94720	2024-04-30	3205	8	92	3205	2019
102	Tilden Room	Yes	48	2495 Bancroft Way, Berkeley, CA 94720	2024-01-16	3205	8	97	3205	2968
78	Anna Head Alumnae Room	Yes	800	2537 Haste St, Berkeley, CA 94720	2024-04-30	3173	7	72	3173	2149
77	Anna Head Alumnae Room	Yes	379	2537 Haste St, Berkeley, CA 94720	2024-02-10	3157	6	86	3157	1336
78	Anna Head Alumnae Room	Yes	800	2537 Haste St, Berkeley, CA 94720	2024-01-29	3142	5	85	3142	1477
78	Anna Head Alumnae Room	Yes	800	2537 Haste St, Berkeley, CA 94720	2024-04-23	3142	8	71	3142	1195
102	Tilden Room	Yes	48	2495 Bancroft Way, Berkeley, CA 94720	2024-07-23	3094	8	89	3094	1495
77	Anna Head Alumnae Room	Yes	379	2537 Haste St, Berkeley, CA 94720	2024-01-17	3088	8	74	3088	1424
78	Anna Head Alumnae Room	Yes	800	2537 Haste St, Berkeley, CA 94720	2024-11-20	3051	6	76	3051	2032
234	Sequoia at Anchor House	No	72	1950 Oxford St, Berkeley, CA 94704	2024-09-05	3050	7	93	3050	1402
78	Anna Head Alumnae Room	Yes	800	2537 Haste St, Berkeley, CA 94720	2024-05-21	3013	5	89	3013	1466

QUERY 1: METHODOLOGY

UC Berkeley Event Services

Demand Forecasting and Profitability Optimization

1. Data cleaning and processing
2. Feature engineering
   a. Removal of outliers
   b. One-hot-encoding
3. Data visualization analysis
4. Building a linear model and feature selection
5. Defining evaluation metrics
   a. Out of Sample R-Squared
   b. Mean Squared Error

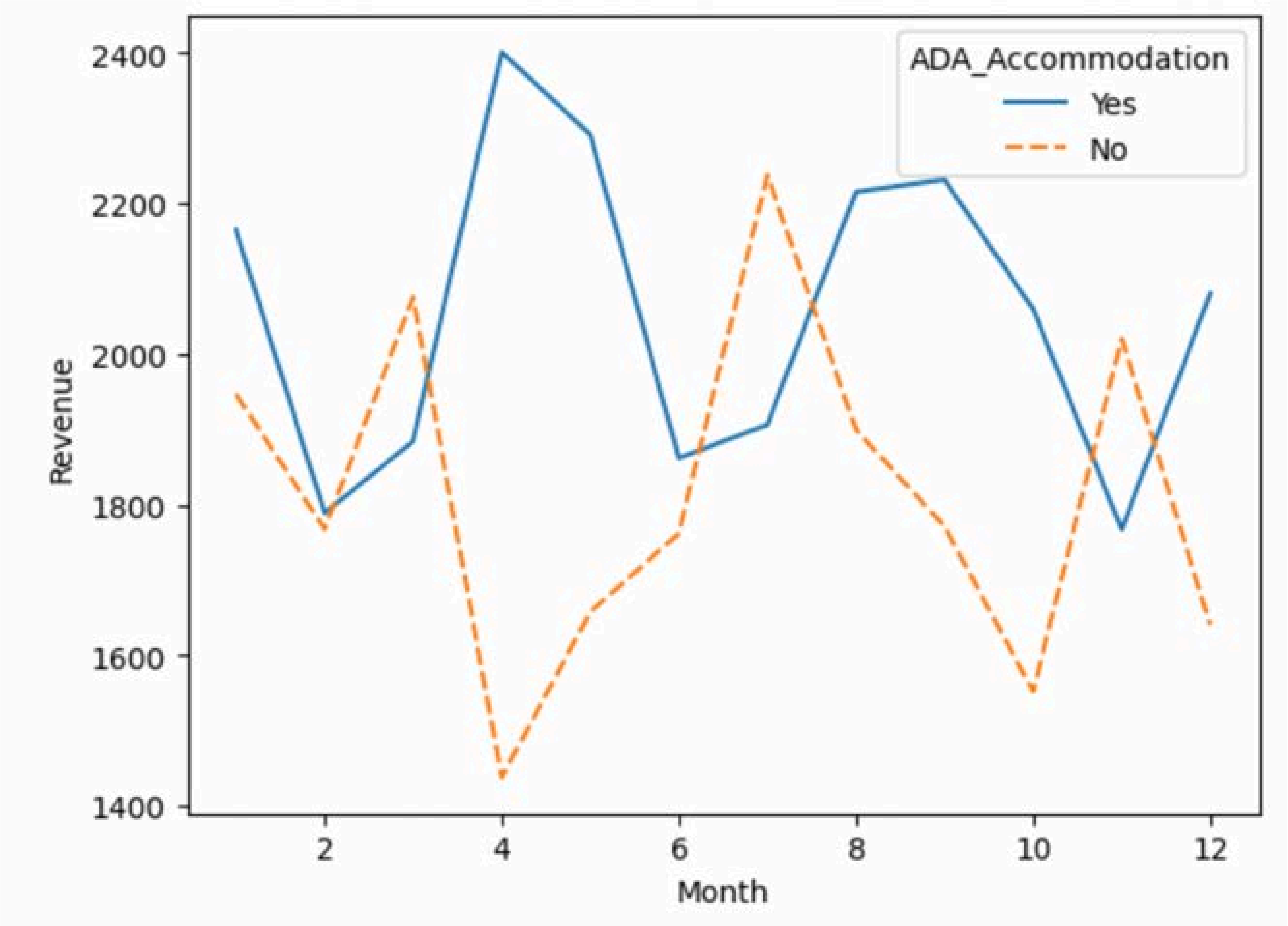
```
# Pre-process data: convert Booking_Date to datetime
query1['Month'] = query1['Booking_Date'].dt.month
query1['Day'] = query1['Booking_Date'].dt.day
query1 = query1.drop('Price Charged',axis=1)
query1.head(1)
```

```
# Define OSR2 and MAE functions
from sklearn.metrics import mean_absolute_error as mae

def OSR2(model, X_test, y_test, y_train):

    y_pred = model.predict(X_test)
    SSE = np.sum((y_test - y_pred)**2)
    SST = np.sum((y_test - np.mean(y_train))**2)

    return (1 - SSE/SST)
```

QUERY 1: EXPLORATORY DATA ANALYSIS

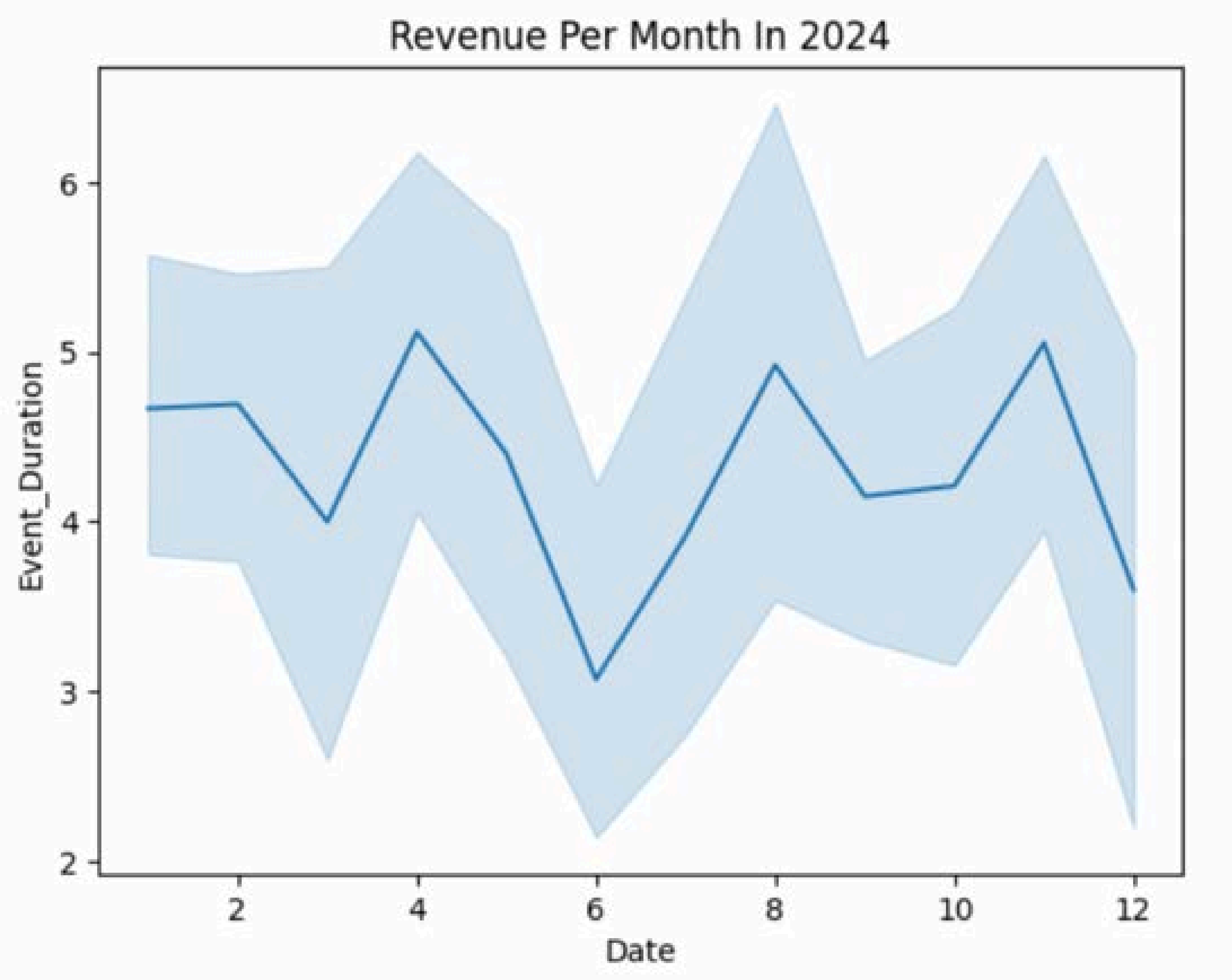
UC Berkeley Event Services

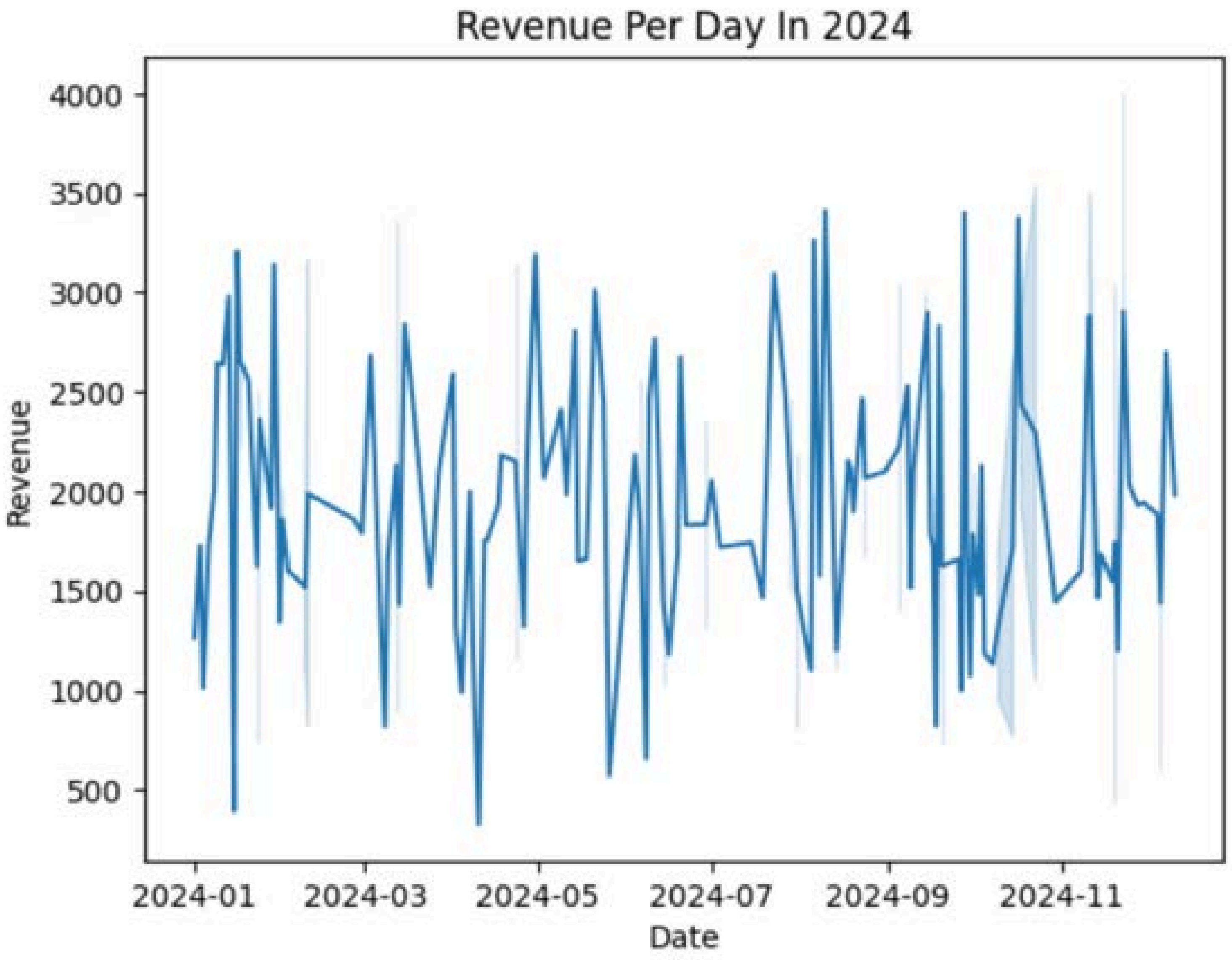
Demand Forecasting and Profitability Optimization



QUERY 1: REVENUE OVER TIME

Demand Forecasting and Profitability Optimization





Observing trends throughout the past year, we can see that there is less revenue generated during the summer and late winter months due to academic breaks.

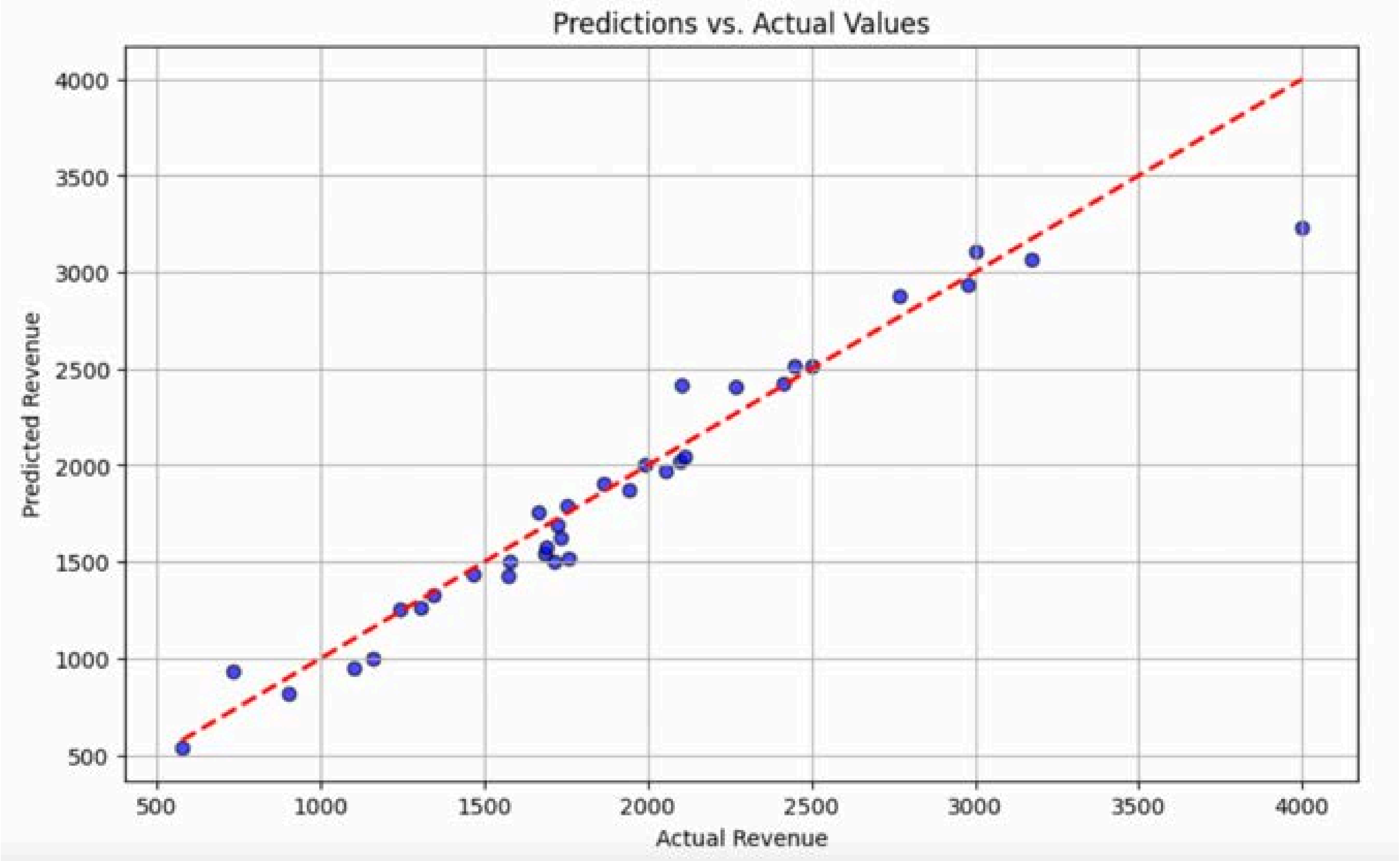
QUERY 1: PREDICTING REVENUE

Demand Forecasting and Profitability Optimization

```
formula="Revenue ~ Capacity + Event_Duration + Labor_Hours + Month"
```

Generated data on revenue for an event was correlated to guest capacity, duration, and labor hours required.

Resulting revenue projection model was highly accurate and can inform future price models that depends on seasonal trends.



R^2: 91.7%

OSR^2: 89.5%

Query 2:

Inventory Depreciation & Replacement Forecasting

QUERY 2: BACKGROUND AND BENEFITS

Lifestyle Tracking

1. Comprehensive **monitoring of equipment** from purchase to retirement

Proactive Planning

2. Anticipating **maintenance** and replacement needs before they become urgent

Financial Optimization

3. Aligning equipment management with **strategic budget planning**

QUERY 2: SQL CODE

Inventory Depreciation & Replacement Forecasting

```
WITH Inventory_Lifetime AS (
    SELECT i_item.SKU, i_item.Item_no, i_item.Purchase_price, i_item.Purchase_date, i_item.Gets_stored_Space_ID,
        i_item.Inv_Is_of_Sale_no, i_item.Inv_Is_for_Order_no, i_type.Type_Name, i_type.Inventory_expected_lifetime,
        TIMESTAMPDIFF(MONTH, i_item.Purchase_date, CURRENT_DATE()) AS Months_since_purchase,
        DATE_ADD(i_item.Purchase_date, INTERVAL (i_type.Inventory_expected_lifetime * 0.9) MONTH) AS Expected_replacement_date,
        ROUND(LEAST(GREATEST(TIMESTAMPDIFF(MONTH, i_item.Purchase_date, CURRENT_DATE()), 0) / NULLIF(i_type.Inventory_expected_lifetime, 0) * 100, 100), 2)
        AS Current_wear_percentage,
        GREATEST(i_type.Inventory_expected_lifetime - TIMESTAMPDIFF(MONTH, i_item.Purchase_date, CURRENT_DATE()), 0)
        AS Remaining_life_months
    FROM INVENTORY_ITEM as i_item, INVENTORY_TYPE as i_type
    WHERE i_item.SKU = i_type.SKU),
Budget_Recs AS (
    SELECT SKU, Item_no, Purchase_price, Purchase_date, Type_Name, Current_wear_percentage, Remaining_life_months, Expected_replacement_date,
        CASE WHEN Remaining_life_months <= 3
            THEN Purchase_price * 1.03
            ELSE 0
        END AS Suggested_replacement_budget
FROM Inventory_Lifetime)
SELECT *
FROM Budget_Recs
ORDER BY Expected_replacement_date;
```

QUERY 2: QUERY RESULTS

Inventory Depreciation & Replacement Forecasting

SKU	Item_no	Purchase_price	Purchase_date	Type_Name	Current_wear_percentage	Remaining_life_months	Expected_replacement_date	Suggested_replacement_budget
44006	1000	500	8/17/08	Chairs	100	0	12/17/10	515
44467	1025	900	1/8/20	Flatscreen HD TV	100	0	2/8/21	927
44162	1011	569	7/5/22	Small Cocktail table poles	100	0	9/5/22	586.07
44138	1010	91	1/3/23	Tall Cocktail table poles	100	0	3/3/23	93.73
44718	1024	879	1/10/23	Fire heater	100	0	3/10/23	905.37
44498	1018	968	5/9/22	8ft classroom tables	100	0	4/9/23	997.0400000000001
44194	1012	675	3/9/23	Chair carts	100	0	7/9/23	695.25
44123	1009	318	10/17/23	Cocktail table base	100	0	12/17/23	327.54
44332	1016	600	8/5/23	6ft round tables	100	0	2/5/24	618
44241	1028	1500	2/22/21	Wooden Podium	100	0	3/22/24	1545
44375	1030	3000	2/21/21	Stage	100	0	4/21/24	3090
44073	1006	297	11/12/23	Sign holders	100	0	6/12/24	305.91
44275	1013	814	10/18/23	Stanchions	100	0	6/18/24	838.4200000000001
44316	1015	482	6/13/24	5ft round tables	100	0	12/13/24	496.46000000000004
44111	1007	868	2/29/24	Eight footer tables	87.21	2	1/29/25	894.0400000000001
44314	1014	642	4/16/24	Easels	80.27	2	2/16/25	661.26
44468	1017	967	11/26/24	6ft classroom tables	8	7	6/26/25	0
44570	1019	766	7/16/22	Table carts	81.56	7	7/16/25	0
44006	1001	500	6/22/23	Chairs	64.26	11	10/22/25	0
44369	1031	100	1/23/22	Monitor	74.81	12	12/23/25	0

QUERY 2: PYTHON CODE

UC Berkeley Event Services

Inventory Depreciation & Replacement Forecasting

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
query3 = pd.read_csv('./query3.csv')
```

```
X = query3.drop(columns=["Remaining_life_months"])

X_temp = X.copy()

dummies = pd.get_dummies(X_temp[['Type_Name']])

X = pd.concat([X_temp.loc[:, ['Current_wear_percentage', 'Purchase_price', 'Type_Name']], dummies_train], axis=1).drop(columns=['Type_Name'])

y = query3["Remaining_life_months"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestRegressor(random_state=42, n_estimators=100)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(12, 8))
sns.scatterplot(x=y_test, y=y_pred, color="dodgerblue", s=120, alpha=0.8, edgecolor="black", linewidth=0.5)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r-', lw=3, label="Perfect Prediction Line")
plt.title("Actual vs Predicted Remaining Life (Months)", fontsize=20, fontweight="bold", pad=20)
plt.xlabel("Actual Remaining Life (Months)", fontsize=14, labelpad=10)
plt.ylabel("Predicted Remaining Life (Months)", fontsize=14, labelpad=10)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.grid(color='gray', linestyle='--', linewidth=0.5, alpha=0.7)
plt.show()
```

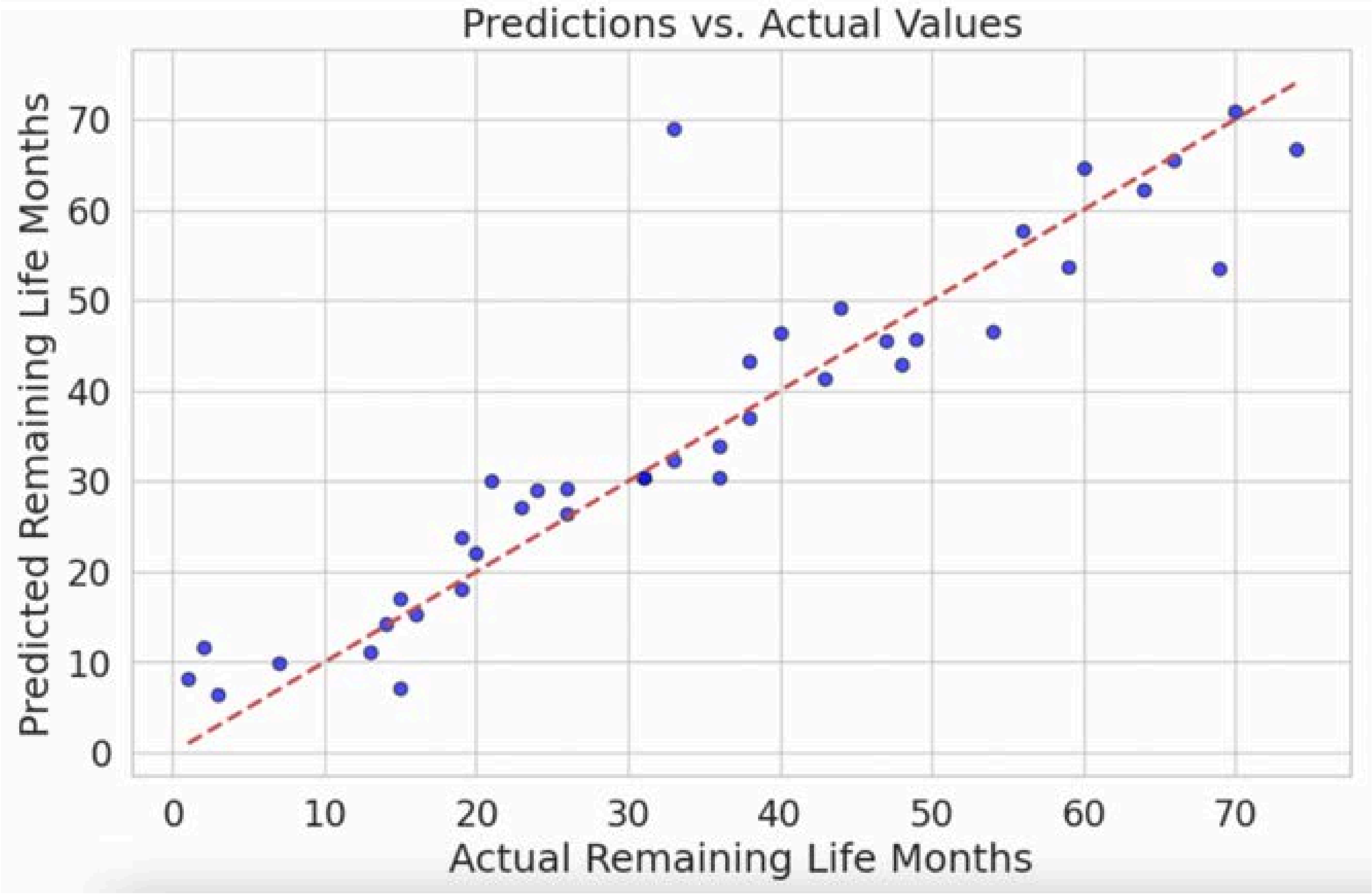
UC Berkeley
Event Services

QUERY 2: REMAINING LIFE ANALYSIS

Inventory Depreciation & Replacement Forecasting

```
# Build regression model
ols = smf.ols(formula="Remaining_life ~ Age + Current_wear_percentage + Purchase_price", data=train).fit()
```

Comparing the actual and predicted remaining life allows us to assess the accuracy of our modeling system and allocate enough budget for upcoming years with large replacement needs

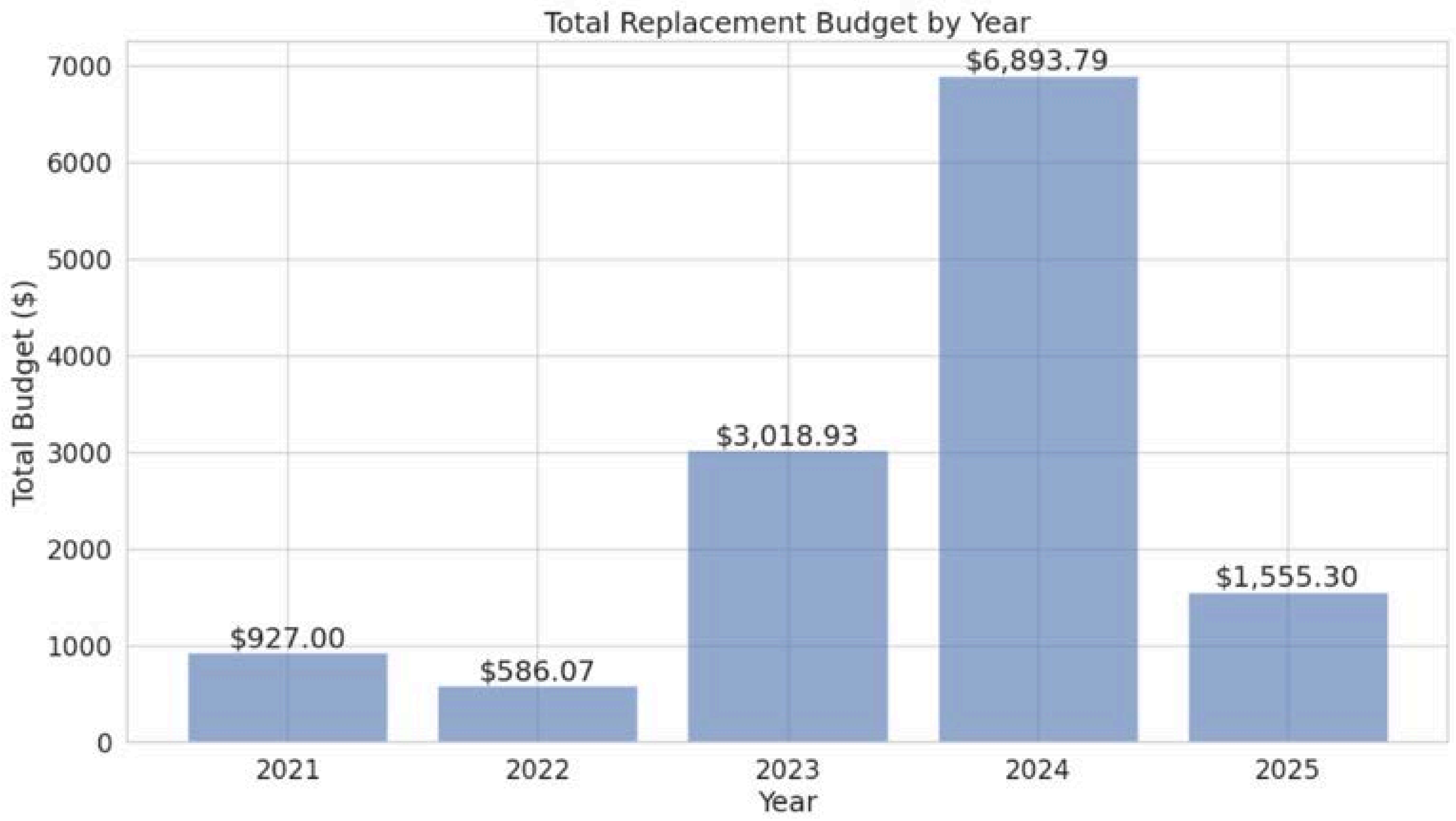


QUERY 2: YEARLY BUDGET ANALYSIS

UC Berkeley Event Services

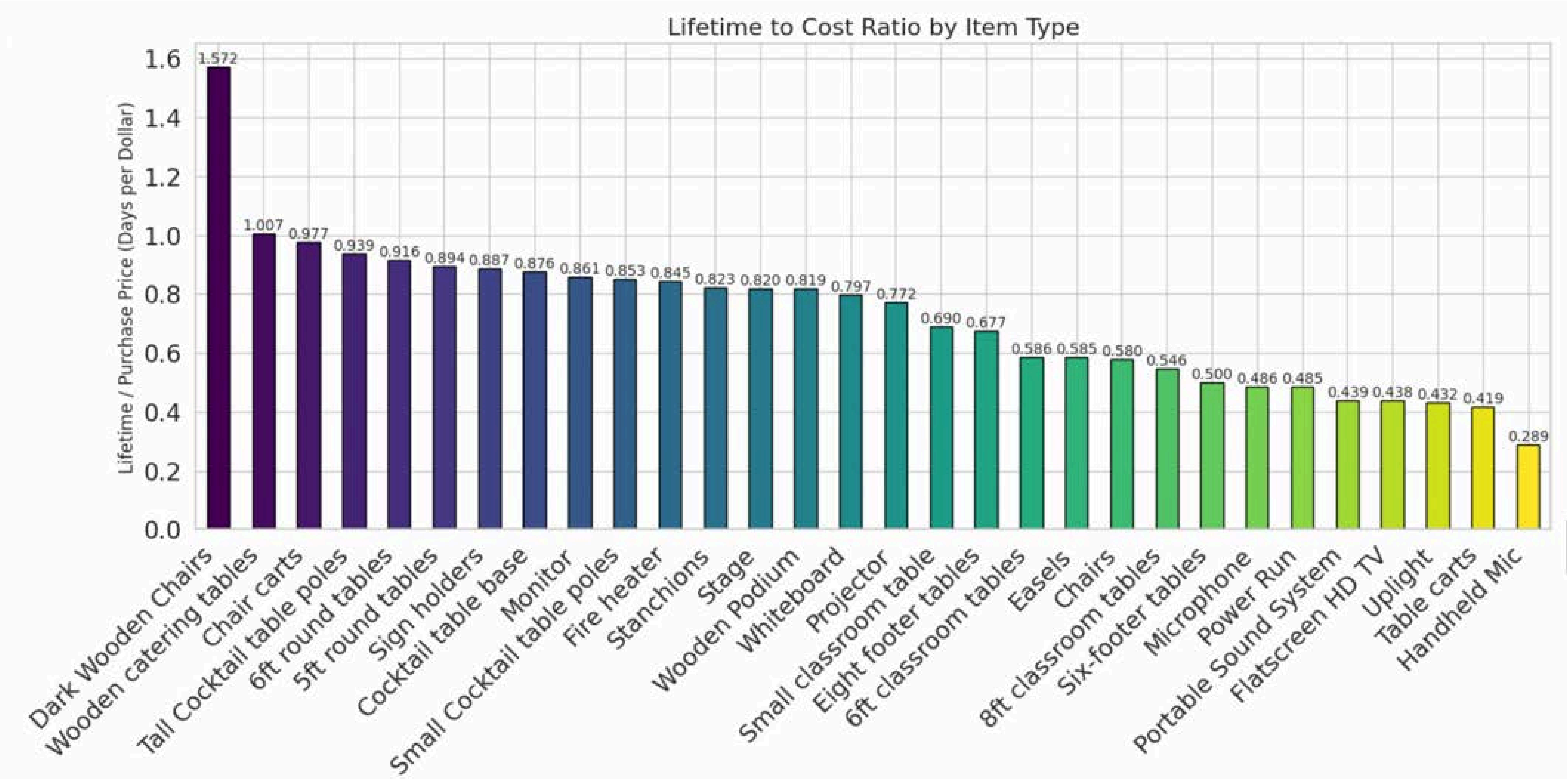
Inventory Depreciation & Replacement Forecasting

Knowing the remaining life of inventory items and resources, Berkeley Event Services can spread out expenses over multiple budgets to reduce financial strain.



QUERY 2: LIFETIME AND COST ANALYSIS

Inventory Depreciation & Replacement Forecasting



Normalization Analysis

UC Berkeley Event Services

UC Berkeley Event Services

FUNCTIONAL DEPENDENCIES

Relations

INVENTORY_TYPE(SKU, Type_name, Inventory_expected_lifetime)

INVENTORY_ITEM(SKU, Item_no, Purchase_price, Purchase_date, | Gets_stored_Space_ID , Inv_Is_of_Sale_no, Inv_Is_for_Order_no)

CLIENT(Client_ID, Client_name, Rate, Mailing_address, | Org_ID)

ORGANIZATION(Org_ID, Num_members, Org_name, Tax_ID)

PERSON(Person_ID, First_name, Last_name)

Functional Dependencies

{SKU} → {Type_name, Inventory_expected_lifetime}

{SKU, Item_no} → {Purchase_price, Purchase_date, Gets_stored_Space_ID, Inv_Is_of_Sale_no, Inv_Is_for_Order_no}

{Client_ID} → {Client_name, Rate, Mailing_address, Org_ID}, {Org_ID}

{Num_members, Org_name, Tax_ID}

{Person_ID} → {First_name, Last_name, Email_addresses}}

NORMALIZATION: 1NF

UC Berkeley Event Services

Relation:
PERSON(Person_ID, First_name, Last_name, Email_addresses)

Functional Dependencies:
{Person_ID} → {First_name, Last_name, Email_addresses}

Normalize PERSON to 1NF because Email_addresses is a multivalued attribute:

1. PERSON(Person_ID, First_name, Last_name)
2. EMAIL_ADDRESSES(Person_ID [1], Email_address)

Also in 2NF and 3NF

NORMALIZATION: 2NF

UC Berkeley Event Services

Relation:
INVENTORY(SKU, Item_no, Type_name, Inventory_expected_lifetime, Purchase_price, Purchase_date, Gets_stored_Space_ID, Inv_Is_of_Sale_no, Inv_Is_for_Order_no)

Functional Dependencies:
{SKU} → {Type_name, Inventory_expected_lifetime}
{SKU, Item_no} → {Purchase_price, Purchase_date, Gets_stored_Space_ID, Inv_Is_of_Sale_no, Inv_Is_for_Order_no}

Normalize INVENTORY from 1NF to 2NF because Inventory_expected_lifetime only depends on SKU and not Item_no:

1. INVENTORY_TYPE(SKU, Inventory_expected_lifetime)
2. INVENTORY_ITEM(SKU[1], Item_no, Quantity, Purchase_price, Purchase_date, Gets_stored_Space_ID, Inv_Is_of_Sale_no, Inv_Is_for_Order_no)

Also in 3NF

NORMALIZATION: 3NF

UC Berkeley Event Services

Relation:
CLIENT(<u>Client_ID</u>, Client_name, Rate, Mailing_address, Org_ID, Num_members, Org_name, Tax_ID)

Functional Dependencies:
{Client_ID} → {Client_name, Rate, Mailing_address, Org_ID}
{Org_ID} → {Num_members, Org_name, Tax_ID}

Normalize CLIENT from 2NF to 3NF because there is a transitive dependency with Org_ID:

1a. ORGANIZATION(<u>Org_ID</u>, Num_members, Org_name, Tax_ID)
1b. CLIENT(<u>Client_ID</u>, Client_name, Rate, Mailing_address, Org_ID[1A])

Also in BCNF

NORMALIZATION: BCNF

UC Berkeley Event Services

Relation:
COORDINATES(Res_ID, Person_ID, Num_Hours_Spent_Coordinating)

Functional Dependencies:
{Res_ID, Person_ID} → {Num_Hours_Spent_Coordinating}

Normalize COORDINATES from 3NF to BCNF because there are functional dependencies that do not have a uniquely identifying key on the left side

1. R1(Person_ID, Num_Hours_Spent_Coordinating)
2. R2(Res_ID, Person_ID[1])

Also in 2NF and 3NF

UC Berkeley Event Services

NORMALIZATION: ANALYSIS

1NF 1. EVENT(Res_ID, Duration, Start_Time, Event_Name, Date, | **Ven_ID[21], Date[21], Reservation_Time[21]**|

BCNF a. UNPAID(Res_ID[1])

- i. PRESENTATION(Res_ID[1a], Presenter_name)
- ii. MEETING(Res_ID[1a], Meeting_topic)
- iii. INTERVIEW(Res_ID[1a], Host_name)

BCNF b. PAID(Res_ID[1], Date_of_payment, Amt_charged, Amt_deposit, Amt_refundable)

- i. CAREER_FAIR_TRADESHOW(Res_ID[1b], Num_of_companies)
- ii. RECEPTION_COCKTAIL(Res_ID[1b], Allergy_accomodation_needed, Host_name)
- iii. BANQUET(Res_ID[1b], Allergy_accomodation_needed)
- iv. LECTURE(Res_ID[1b], Lecturer_name)
- v. WEDDING(Res_ID[1b], Allergy_accomodation_needed)
- vi. FILM_SCREENING(Res_ID[1b], Producer_name)
- vii. DANCE_CONCERT(Res_ID[1b], Performance_type)
- viii. CONFERENCE(Res_ID[1b], Conference_type)

3NF 2. ADDITIONAL_ACCOMODATION(Request_no, Res_ID[1], Request_detail)

BCNF 3. SERVICE(Service_name, Service_type)

BCNF 4. PERSON(Person_ID, First_name, Last_name)

3NF a. EMPLOYEE(Person_ID[4], SSN, Start_date, Mailing_address, Bank_account_no, Birthdate, | **Makes_Budget_ID[19]**)

- i. FULL_TIME(Person_ID[4a], Salary)
- ii. PART_TIME(Person_ID[4a], Hourly_wage)

b. CLIENT_CONTACT(Person_ID[4], Relation_to_client, | **Delegate_of_Client_ID[8]**)

1NF 5. EMAIL_ADDRESSES(Person_ID[4], Email_address)

BCNF 6. LIABILITY_WAVER(Waiver_ID, Date_last_updated)

BCNF 7. LOCATION(Address, Building_name)

3NF 8. CLIENT(Client_ID, Client_name, Rate, Mailing_address, | **Org_ID[11]**)

BCNF 9. INVENTORY_TYPE(SKU, Type_name, Inventory_expected_lifetime)

2NF 10. INVENTORY_ITEM(SKU[9], Item_no, Purchase_price, Purchase_date, | **Gets_stored_Space_ID[20], Inv_Is_of_Sale_no[16], Inv_Is_for_Order_no[17]**)

BCNF 11. ORGANIZATION(Org_ID, Num_members, Org_name, Tax_ID)

a. EXTERNAL_ORG(Org_ID[11], Is_nonprofit)

b. INTERNAL_ORG(Org_ID[11], Has_student_members)

- i. RSO(Org_ID[11b], Advisor_department, Advisor_name)
- ii. CAMPUS_DEPARTMENT(Org_ID[11b], Dept_name)

1NF 12. REQUEST(Request_ID, request_date, | **Made_by_Client_ID[8]**)

BCNF 13. RESOURCE(Resource_name, Resource_expected_lifetime)

- a. TV(Resource_Name[13], Quality, Resolution, Screen_size)
- b. LAV_MIC(Resource_Name[13], Resource_frequency)
- c. PORTABLE_SOUND_SYSTEM(Resource_Name[13], Quality)
- d. PODIUM(Resource_Name[13], Size)
- e. POWER_RUN(Resource_Name[13], Resource_voltage)
- f. STAGE(Resource_Name[13], Size)
- g. MONITOR(Resource_Name[13], Quality, Resolution, Screen_size)
- h. PROJECTOR(Resource_Name[13], Quality, Is_built_in)
- i. HANDHELD_MIC(Resource_Name[13], Resource_frequency)
- j. UPLIGHT(Resource_Name[13], Resource_lumens)

1NF 14. VENDOR(Vendor_name, Mailing_address, Cancellation_refundable)

2NF 15. VENUE(Ven_ID, Venue_name, ADA_accomodation, Capacity, **Is_booked_at_Address[7], Ven_Is_of_Sale_no[16], Ven_Is_for_Order_no[17]**)

- a. LOUNGE(Ven_ID[15], Kitchen_included)
- b. COURTYARD(Ven_ID[15], Weather_accomodation_plan)
- c. THEATER(Ven_ID[15], Orchestra_pit_included, Stage_lights_included)
- d. CONFERENCE_ROOM(Ven_ID[15], Can_livestream)
- e. CLASSROOM(Ven_ID[15], Chalkboard_or_whiteboard)
- f. BALLROOM(Ven_ID[15], Floor_type)

1NF 16. SALE(Sale_no, Dollar_amt, Date_completed)

1NF 17. PURCHASE(Order_no, Dollar_amt, Date_completed)

UC Berkeley Event Services

NORMALIZATION: ANALYSIS

18. ACCOUNT(Account_no, Bank_name) BCNF
    a. RECEIVABLE(Account_no[18], Internal_funds)
    b. PAYABLE(Account_no[18], Reserved_funds)
19. BUDGET(Budget_ID, Dollar_amt, Number_list_items, Date_set, Percent_accurate) 3NF
    a. OPERATING EXPENSE(Budget_ID[19], Intended_start_date, Intended_end_date)
    b. PROFESSIONAL_DEVELOPMENT(Budget_ID[19], Expenditure_date, Start_time, Duration) BCNF
    c. CAPITAL EXPENDITURE(Budget_ID[19], Expenditure_date) BCNF
    d. STAFFING(Budget_ID[19], Intended_start_date, Intended_end_date)
20. STORAGE SPACE(Space_ID, Capacity, | **Is_located_at_Address**[7]) BCNF
21. AVAILABILITY_SCHEDULE(Ven_ID[15], Date, Reservation_Time, AfterHours_or_ActiveHours) 3NF

22. COORDINATES(Res_ID[1], Person_ID[4a], Num_Hours_Spent_Coordinating) BCNF
23. MAINTAINS(Ven_ID[15], Person_ID[4a], Num_Hours_Spent_Maintaining) BCNF
24. FOR_DATES(Request_ID[12], Date[21], Reservation_Time[21]) BCNF
25. I_INCLUDES(Res_ID[1], SKU[9], Num_items) BCNF
26. REQUIRES_SERVICE(Res_ID[1], Service_name[3], Budgeted_amount) BCNF
27. PROVIDED_BY(Service_name[3], Vendor_name[14], Max_num_vendors) 3NF
28. CONTRACTS(Person_ID[4b], Vendor_name[14], Payment_date) BCNF
29. MAKES_RSO(Res_ID[1a], Org_ID[11bi], Affiliated_with) BCNF
30. PAYS(Client_ID[8], Account_no[18a], Amount) BCNF
31. FUNDED_BY(Budget_ID[19], Account_no[18b], Transfer_date, Funding_date) BCNF
32. FUNDS(Sale_no[16], Account_no[18a], Amount) BCNF
33. GETS_PAID(Person_ID[4a], Budget_ID[19d], Amount) BCNF
34. CONFIRMS(Person_ID[4a], Order_no[17], Purchase_date) BCNF
35. APPROVES(Person_ID[4a], Sale_no[16], Sale_date) BCNF
36. GRANTS(Order_no[17], Budget_ID[19c], Amount, Reason) 3NF
37. CONTAINS(Ven_ID[15], Resource_Name[13], Total_number_of_items) 3NF
38. COMPENSATES(Budget_ID[19a], Vendor_name[14], Amount) 3NF
39. OF_TYPE(Request_ID[12], Resource_Name[13], Quantity) BCNF
40. REQUIRES_WAIVER(Waiver_ID[6], Res_ID[1b], Date_signed) BCNF
41. ALLOCATED_TO(Ven_ID[21], Date[21], Reservation_Time[21], Resource_Name, quantity_requested) BCNF

UC Berkeley
Event Services

TEAM CONTRIBUTIONS

Note: Contributions may not be exhaustive, as every task was a collaborative effort in some capacity.

- **Andrew:** Helped create EER diagram and relational schema, helped with generating realistic data, that client could use, contributed to creating slide deck,
- **Ashmita:** MySQL Workbench creation, connections based on relational schema, generation of data, running SQL code, created visualization and models for Query 2 in Python
- **Christopher:** Helped with creating EER diagram, worked on slide deck for DP reviews, helped generate data, work on Python code for queries and models/visualizations
- **Ella:** MySQL Workbench creation, relational schema creation, data generation, running SQL code, EER diagram creation and updates, team coordination and task delegation
- **Ernest:** Created and updated slide decks for the DP reviews, helped with creating query background, assisted with EER diagram updates
- **Gursimar:** Assisted with EER diagram creation, created and updated slide decks for the DP reviews, helped with creating query descriptions/background, created SQL code for Query 1, 2 and SQL output generation, generated simulated data for few tables

UC Berkeley
Event Services

TEAM CONTRIBUTIONS

- **Hannah:** Normalization analysis, created visualizations and models for Query 1 in Python, made Query 1 slides, assisted with EER diagram updates, redesigned slide deck into final version
- **Jacqueline:** My SQL Workbench creation and relational schema connections, generated simulated data, EER diagram creation and updates, helped with logical formulating SQL background and code for queries 1 & 2, updated slides, team coordination
- **Joanne:** Client contact, EER diagram creation, created SQL Query 2 and ran both 1 and 2 on MySQL to generate CSV tables, random data generation
- **Michael:** Created visualizations for Query 3 in Python, created SQL Query 1, ideation of queries, slide deck design
- **Moritz:** EER diagram design and updates, query generation and analysis, slide deck design
- **Shishira:** Normalization analysis (BCNF), generated data with PyGen for both queries, assisted with EER diagram updates, redesigned slide deck into final version